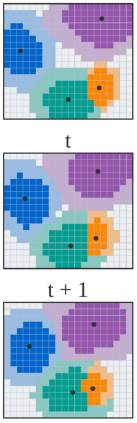
Input

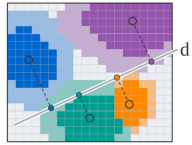
$F(x,t)$, budget ε



scalar fields over time, one budget

(1) Trees and reference

(a) Spatial projection



(b) Merge tree T_t



merge trees T_t, matched supports, one direction a^*, references q

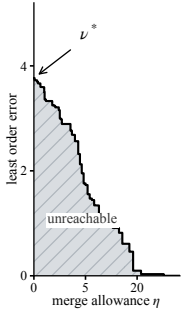
(2) Price of the axis

from T, q, W only

$\nu^*(T,q)$ exact tree

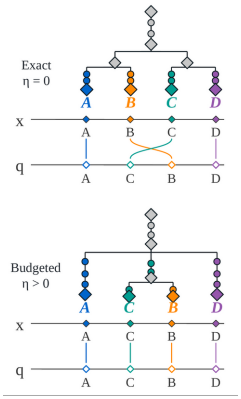
$\mathcal{B}_w(T,q,W)$ finite widths

$C_\eta(T,q)$ merge error $\le \eta$



known before layout: how much the tree forces, where, and what merge error releases it

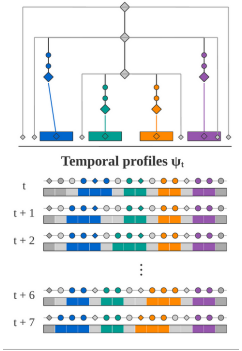
(3) Leaf order and anchors



admit π if $\delta^*(\pi) \le \eta$ (closed form)
choose least $\tau(\pi)$, then NROE

anchors: convex QP with $\|x-q\|_\infty \le \tau(\pi) + \beta_0$

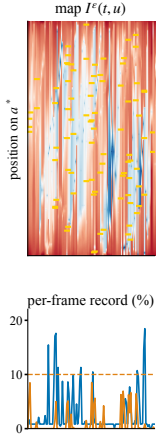
(4) Profiles and checks



barriers $b_k = C_k + \delta^*(\pi)$
check $d_{\text{top}} \le \eta$ on every frame

profiles ψ_t share origin, direction and scale

Output



barriers changed (yellow); displacement (blue), merge error within ε